

Defⁿ A family $f(x|\theta) = h(x) e^{\sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta)}$

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Para Inf

(Similarly $f(x|\eta) = h(x) \exp(\sum \eta_i T_i(x) - A(\eta))$)

is called a k -para "full rank" exp family
if the representation is minimal and the natural
parameter space $\mathcal{N}$ contains an open rectangle in $\mathbb{R}^k$

Thm If $X = (X_1, \dots, X_n)$ has pmf/pdf $f(x|\theta) = h(x) e^{\sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta)}$
that belongs to a k -parameter full rank exponential family,
then $T(X) = (T_1(X), \dots, T_k(X))$ is a complete and sufficient
statistic for θ

Pf ($k=1$, X discrete)

Given $f_X(x|\theta) = h(x) e^{\eta(\theta) T(x) - B(\theta)} \in$ full rank exp.
family

TS $T(X)$ is complete

TS If for some g : $E_\theta[g(T)] = 0 \forall \theta$
 $\Rightarrow P_\theta[g(T) = 0] = 1 \forall \theta$

The pmf for T is $P_\theta[T(X) = t] = f_T(t|\theta) = h^*(t) e^{\eta(\theta)t - B(\theta)}$
 $= e^{\eta(\theta)t + S^*(t) - B(\theta)}$
for $t \in \text{supp}(T)$

where $S^*(t) = \ln(h^*(t))$

Given: $E_\theta[g(T)]$

$$= \sum_t g(t) e^{\eta(\theta)t + S^*(t) - B(\theta)} = 0 \quad \forall \theta \in \Theta$$

" $g^+(t) - g^-(t)$

$$\Rightarrow \text{Suppose } \sum_t g^+(t) e^{\eta(\theta)t + S^*(t)} = \sum_t g^-(t) e^{\eta(\theta)t + S^*(t)} \quad \forall \theta \in \Theta$$
$$= c(\theta) \quad (\text{say})$$

Suppose $(\alpha, \beta) \subset \mathcal{N}$. Let $\theta_0 \in \Theta$ s.t. $\eta(\theta_0) \in (\alpha, \beta)$

Now define $p^+(t) = \frac{g^+(t) e^{\eta(\theta_0)t + S^*(t)}}{c(\theta_0)}$ is indeed a pmf

$p^-(t) = \frac{g^-(t) e^{\eta(\theta_0)t + S^*(t)}}{c(\theta_0)}$ is a pmf as well

MGF $\sum_t e^{\delta t} p^+(t) = \frac{1}{c(\theta_0)} \sum_t g^+(t) e^{(\eta(\theta_0) + \delta)t + S^*(t)} < \infty$

exists $\forall \delta \in (\alpha - \eta(\theta_0), \beta - \eta(\theta_0))$

Similarly $\sum_t e^{\delta t} p^-(t) < \infty \quad \forall \delta \in (\alpha - \eta(\theta_0), \beta - \eta(\theta_0))$

$$\sum_t e^{\delta t} p^+(t) = \sum_t e^{\delta t} p^-(t) \text{ hold for } \forall \delta \in (\alpha - \eta(\theta_0), \beta - \eta(\theta_0))$$

Uniqueness of MGF, we get $p^+(t) = p^-(t) \quad \forall t \in \text{supp}(T)$

$$\Rightarrow q^+(t) = q^-(t) \quad \forall t \in \text{supp}(T)$$

$$\Rightarrow q(t) = 0 \quad \forall t \in \text{supp}(T)$$

$$\Rightarrow P_\theta(q(T)=0) = 1 \quad \forall \theta \in \Theta$$

Ex. $X_1, \dots, X_n$ iid $\text{Gamma}(\alpha, \beta)$ $\alpha > 0$
 $\beta > 0$
 Find C.S.S. for $\theta = (\alpha, \beta)$

$$f(x|\theta) = \frac{e^{-x/\beta} x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)}, \quad x > 0 \in \text{full rank 2-param exp family}$$

$$= x^{-1} \exp \left\{ -\frac{x}{\beta} + \alpha \ln x + \ln(\beta^\alpha \Gamma(\alpha)) \right\}$$

$$T_1(x) = x \quad \eta_1(\theta) = -1/\beta$$

$$T_2(x) = \ln x \quad \eta_2(\theta) = \alpha$$

$$\mathcal{N} = \left\{ (-1/\beta, \alpha) : \alpha > 0, \beta > 0 \right\} = \mathbb{R}^- \times \mathbb{R}^+ \text{ does contain}$$

an open rectangle in $\mathbb{R}^2$

$$\text{Complete suff stat for } \theta \text{ is } T(\underline{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n \ln X_i \right)$$

Ex. $\mathcal{N}(\mu, \sigma^2)$, $\text{Beta}(\alpha, \beta)$, $\text{Bin}(k, \theta)$, $\text{Pois}(\lambda)$, $\text{Geo}(\theta)$

Properties of Exponential Family:

Suppose the canonical form of a family of distⁿ is $f(x|\eta) = h(x) e^{\sum_{i=1}^k \eta_i T_i(x) - A(\eta)}$. For any integrable function $g(x)$ and η in the interior of $\mathcal{N}$, we have

(1) $\int g(x) h(x) e^{\sum_{i=1}^k \eta_i T_i(x) - A(\eta)} dx$ is cont. in η_i and has derivatives in η of all orders $\forall i=1, \dots, k$

$$(2) \quad \frac{\partial}{\partial \eta_i} \int g(x) h(x) e^{\sum_{i=1}^k \eta_i T_i(x) - A(\eta)} dx \\ = \int \frac{\partial}{\partial \eta_i} g(x) h(x) e^{\sum_{i=1}^k \eta_i T_i(x) - A(\eta)} dx$$

(3) MGF of $T(x) = (T_1(x), \dots, T_k(x))$ exists in some nbhd. of 0, and given $M_T(\underline{u}) = E(e^{\underline{u}' T}) = \frac{e^{A(\underline{u}+\eta)}}{e^{A(\eta)}}$

Application $\frac{\partial}{\partial \eta_i} \int h(x) e^{\sum \eta_i T_i(x) - A(\eta)} dx$

" $= \frac{\partial}{\partial \eta_i} (1) = 0$

$$0 = \int h(x) e^{\sum \eta_i T_i(x) - A(\eta)} \left[T_i(x) - \frac{\partial}{\partial \eta_i} A(\eta) \right] dx$$

$$\Rightarrow E T_i(X) = \frac{\partial}{\partial \eta_i} A(\eta) \longrightarrow (\text{Assume derivative exists})$$

Show that, $\frac{\partial^2}{\partial \eta_i \partial \eta_j} A(\eta) = \text{Cov}(T_i(X), T_j(X))$

Application $\underline{X} = (X_1, \dots, X_k) \sim \text{Multinomial}(n, p_1, \dots, p_k)$

$$\mathbb{E} X_i = np_i$$

$$\text{Cov}(X_i, X_j) = \begin{cases} -np_i p_j & \text{if } i \neq j \\ np_i(1-p_i) & \text{if } i = j \end{cases}$$

$$\sum_{i=1}^k p_i = 1$$

$$\sum_{i=1}^k n p_i = n$$

Show this using
the above result

Evaluation of a point estimator :

Suppose $\delta(\underline{X})$ is a point estimator of some fn of θ , $\psi(\theta)$

We say δ is a "good" estimator of ψ if δ is "close enough" to ψ

$\psi : \Theta \rightarrow \mathbb{R}^k$
 $\phi : \mathcal{X} \rightarrow \mathbb{R}^k$
 we can discuss for $k=1$

Measures of closeness : Will say $\delta(\underline{X})$ is good if

(1) $\mathbb{E}(|\delta(\underline{X}) - \psi(\theta)|^r)$ is small for $r \geq 1$

(2) Fix $\epsilon > 0$ $\mathbb{P}[|\delta(\underline{X}) - \psi(\theta)| > \epsilon]$ small

$\longrightarrow$ large-sample property

(2) large sample property — consistency

We say $\delta(X) \equiv \delta_n(X)$ is a consistent estimator
if $\forall \epsilon > 0 \quad \mathbb{P}[|\delta_n(X) - \psi(\theta)| > \epsilon] \rightarrow 0 \text{ as } n \rightarrow \infty$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bin}(K, \theta)$ Find consistent est.
for K .

$$\begin{aligned} X_{(n)} \text{ is consistent } & \mathbb{P}[|X_{(n)} - K| > \epsilon] \\ \text{Fix } 0 < \epsilon < 1 & = \mathbb{P}[X_{(n)} < K - \epsilon] \\ & = (\mathbb{P}[X_1 < K - \epsilon])^n \\ & = (1 - \mathbb{P}[X_1 \geq K - \epsilon])^n \\ & = (1 - \mathbb{P}[X_1 = K])^n \\ & = (1 - \theta^K)^n \rightarrow 0 \text{ as } n \rightarrow \infty \end{aligned}$$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta) ; X_{(n)} \xrightarrow{P} \theta$

Result: $X_1, \dots, X_n \stackrel{iid}{\sim} f_X$ — pdf of X_i
 ξ_p = population quantile of $X_i = F^{-1}(p)$
 $\hat{\xi}_p$ = sample quantile = $X_{[np]}$

$$\textcircled{1} \quad \sqrt{n}(\hat{\xi}_P - \xi_P) \xrightarrow{d} N\left(0, \frac{P(1-P)}{[f_X(\xi_P)]^2}\right) \text{ provided } f_X(\xi_P)^{-1} < \infty$$

$$\textcircled{2} \quad E\hat{\xi}_P = \xi_P + O\left(\frac{1}{n}\right)$$

$$\& \quad V\hat{\xi}_P = \frac{P(1-P)}{n[f_X(\xi_P)]^2} + O\left(\frac{1}{n^2}\right)$$

If δ_n s.t. $E\delta_n \rightarrow \psi(\theta)$ and $V(\delta_n) \rightarrow 0$

$$\Rightarrow \delta_n \xrightarrow{P} \psi(\theta)$$

(by Chebyshev)

$$\Rightarrow \hat{\xi}_P \xrightarrow{P} \xi_P$$

Ex. Sample raw moments are consistent estimators of pop^l raw moment (if sample is iid)

Criteria $E|\delta(X) - \psi(\theta)|^r$ small

The popular choice of $r=2$

$$\begin{aligned} \text{Mean Square Error, } MSE_{\theta}(\delta) &= E_{\theta}(|\delta(X) - \psi(\theta)|^2) \\ &= V_{\theta}(\delta(X)) + \text{Bias}^2(\delta(X), \psi(\theta)) \end{aligned}$$

$$\text{Bias}(\delta(X), \psi(\theta)) = \mathbb{E}_{\theta}(\delta(X) - \psi(\theta))$$

Small sample property

$$\text{Want: } \text{Bias}(\delta(X), \psi(\theta)) = 0$$

$$\mathbb{E}(\delta(X) - \psi(\theta)) > 0 \longrightarrow \text{overestimation}$$

$$\mathbb{E}(\delta(X) - \psi(\theta)) < 0 \longrightarrow \text{underestimation}$$

• $\delta(X)$ is an unbiased estimator of $\psi(\theta)$ if

$$\mathbb{E}_{\theta}(\delta(X)) = \psi(\theta) \quad \forall \theta \in \Theta$$

• If $\exists$ an u.e. of $\psi(\theta)$, then $\psi(\theta)$ is called "estimate".

Ex. $X \sim \text{Bern}(\theta)$ Find u.e. of $\psi(\theta) = \theta^2$

$$\text{If } \mathbb{E}[\delta(X)] = \theta^2$$

$$\Rightarrow (1-\theta)\delta(0) + \theta\delta(1) = \theta^2 \quad \forall 0 < \theta < 1$$

which is not possible

$X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Bern}(\theta)$. Find u.e. of $\psi(\theta) = \theta^2$

$$\delta(X) = X_1 X_2$$

$X_1, X_2, \dots, X_r \stackrel{\text{iid}}{\sim} \text{Bern}(\theta)$ Is $\psi(\theta) = \theta^r$ estimable

Show that $T = \sum_{i=1}^r X_i$, $\frac{(T)_r}{(n)_r}$ u.e. of θ^r if $r \in \mathbb{N}, r < n$

Ex: $X \sim \text{Pois}(\lambda)$ Show $(-2)^X$ is u.e. of $e^{-3\lambda}$
Absurd u.e.